Functions: Composite & Inverse

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

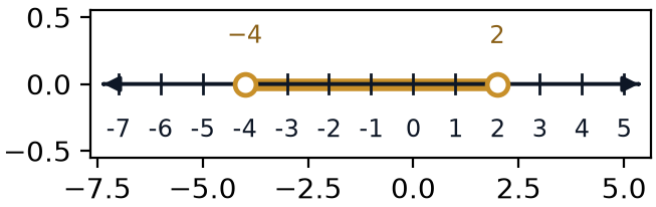
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A vertical pole 6 m tall casts a shadow 9 m long. Find the angle of elevation of the sun, to the nearest tenth of a degree.

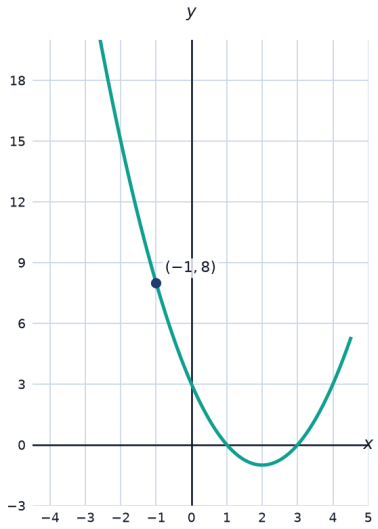
2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve $|2x - 4| < 2$. What is the upper bound of the solution interval? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

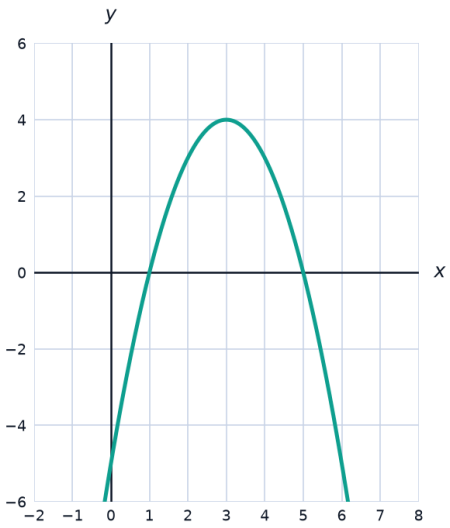
Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.



© ClipSAT — clipsat.org | All Rights Reserved

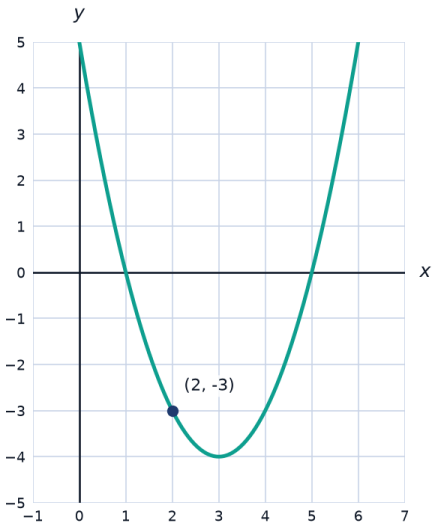
4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.



5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

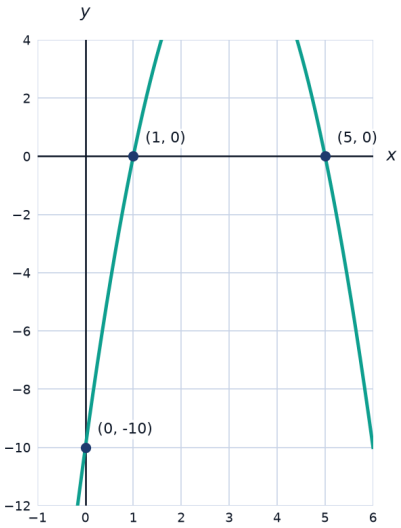
The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(2, -3)$. Determine the value of k.



© ClipSAT — clipsat.org | All Rights Reserved

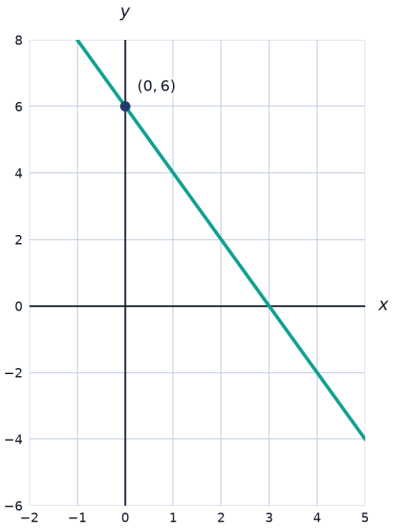
6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(1, 0)$, $(5, 0)$, and $(0, -10)$. Find the value of a.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the y-intercept.

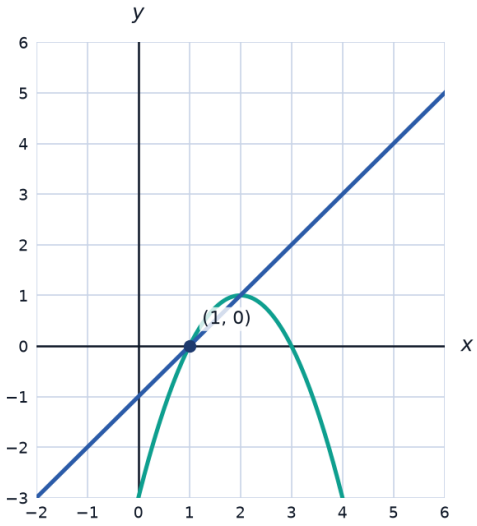


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept? Show your complete algebraic work.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the set of values of the constant m for which the equation $|2x - 6| = mx + 2$ has exactly two distinct real solutions.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the set of values of the constant k for which the equation $|x^2 - 4| = 2x + k$ has exactly four distinct real solutions.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The functions f and g are defined for all real values of x by $f(x) = 2x - 1$ and $g(x) = \frac{x+5}{3}$. Find the value of x such that $fg(x) = 9$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$ has real roots α and β. Given that $\alpha^2 + \beta^2 = 27$, find the value of the constant k.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The line $y = mx - 1$ intersects the curve $y = x^2 + 3x + k$ at two distinct points for all real values of m. Find the range of values of the constant k.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Trigonometry

The function f is defined for $x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Find an expression for $f^{-1}(x)$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Differentiation & Integration

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x \to \infty} f(x) = \frac{1}{2}$ and $\lim_{x \to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3? Show your complete algebraic work.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Circular Measure

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2? Show your complete algebraic work.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The graph of the quadratic function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). Find the value of k.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the natural domain and range of the function $f(x) = \sqrt{2x - 6}$.

a. Domain: $x \geq 3$; Range: $f(x) \geq 0$

b. Domain: $x > 3$; Range: $f(x) > 0$

c. Domain: $x \geq 0$; Range: $f(x) \geq -6$

d. Domain: $x \in \mathbb{R}$; Range: $f(x) \geq 0$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $g(x) = x^2 - 4x + 7$ is defined for $x \in \mathbb{R}$. What is the range of g?

a. $g(x) \geq 3$

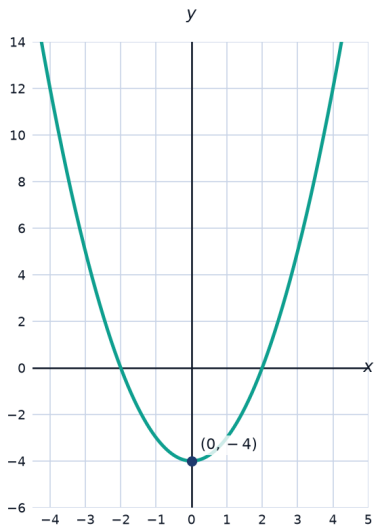
b. $g(x) \geq 7$

c. $g(x) \geq -3$

d. $g(x) \in \mathbb{R}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Explain why the function $h(x) = x^2 - 4$ for $x \in \mathbb{R}$ does not have an inverse.



a. It is a many-to-one function on $\mathbb{R}$

b. Its domain does not include negative numbers

c. Its range is bounded below by -4

d. It is not continuous at $x = 0$

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find the expression for $fg(x)$.

a. $2x^2 + 1$

b. $4x^2 + 12x + 8$

c. $2x^2 + 5$

d. $2x^2 - 2$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $f(x) = \frac{1}{x+2}$ for $x \neq -2$ and $g(x) = 3x - 1$, solve the equation $gf(x) = 1$.

a. $x = 1/2$

b. $x = -1$

c. $x = 2$

d. $x = 0$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $f(x) = 3x - 4$, find the expression for $ff(x)$.

a. $9x - 16$

b. $9x - 12$

c. $6x - 8$

d. $9x^2 - 24x + 16$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x-5}{x+1}$ $(x \neq -1)$.

a. $f^{-1}(x) = \frac{x+5}{2-x}$

b. $f^{-1}(x) = \frac{x-5}{2-x}$

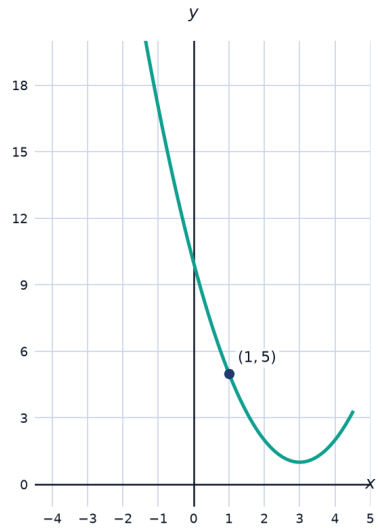
c. $f^{-1}(x) = \frac{2x+1}{x-5}$

d. $f^{-1}(x) = \frac{x+1}{2x-5}$

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function $f(x) = x^2 - 6x + 10$ is defined for $x \geq 3$. Find the expression for $f^{-1}(x)$.



a. $f^{-1}(x) = 3 + \sqrt{x - 1}$ for $x \geq 1$

b. $f^{-1}(x) = 3 - \sqrt{x - 1}$ for $x \geq 1$

c. $f^{-1}(x) = \sqrt{x + 6} - 3$

d. $f^{-1}(x) = 3 + \sqrt{x + 1}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A function $g(x) = \frac{ax + b}{cx + d}$ is a self-inverse function ($g(x) = g^{-1}(x)$). Which condition guarantees this property for all valid x?

a. $a + d = 0$

b. $ad - bc = 1$

c. $a = d$

d. $b = c$

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the geometric relationship between the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$?

a. Reflection in the line $y = x$

b. Reflection in the x-axis

c. Reflection in the y-axis

d. Rotation of 180° about the origin

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the equation $f(x) = f^{-1}(x)$ where $f(x) = 2x - 3$.

a. $x = 3$

b. $x = 1$

c. $x = -3$

d. $x = 2$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function $f(x) = 2x^2 - 8x + c$ has domain $x \geq k$. Find the least value of k for which f has an inverse.

a. $k = 2$

b. $k = 4$

c. $k = 0$

d. $k = -2$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

a. $x = 0$ and $x = 4$

b. $x = 0$ and $x = -4$

c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

© ClipSAT — clipsat.org | All Rights Reserved

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The function f is defined for $x \in \mathbb{R}$ by $f(x) = x^2 - 2kx + (3k + 10)$, where k is a real constant, and the equation $f(x) = 0$ has no real roots. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - k|$. Find the set of values of k for which the minimum value of the composite function $h(x) = f(g(x))$ is strictly greater than 4.

a. $-2 < k < \frac{3+\sqrt{33}}{2}$

b. $\frac{3-\sqrt{33}}{2} < k < \frac{3+\sqrt{33}}{2}$

c. $-2 < k < 5$

d. $0 \le k < \frac{3+\sqrt{33}}{2}$

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The function f is defined by $f(x) = \frac{2x-1}{x+3}$, for $x \neq -3$. Which of the following expressions represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{2-x}, \quad x \neq 2$

b. $f^{-1}(x) = \frac{x+3}{2x-1}, \quad x \neq \frac{1}{2}$

c. $f^{-1}(x) = \frac{3x-1}{x-2}, \quad x \neq 2$

d. $f^{-1}(x) = \frac{2x+1}{x-3}, \quad x \neq 3$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $xe2$. Find the inverse function $f^{-1}(x)$.

a. $f^{-1}(x) = \frac{3x-2}{x+1}$

b. $f^{-1}(x) = \frac{2x-1}{x-3}$

c. $f^{-1}(x) = \frac{x-2}{3x+1}$

d. $f^{-1}(x) = \frac{2x+1}{x-3} \quad (xe3)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1-x^2}$

b. $\frac{1}{\sqrt{1-x^2}}$

c. $1-x^2$

d. $\sqrt{x^2-1}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Circular Measure

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

c. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Series & Binomial Theorem

A 13 m ladder leans against a vertical wall and reaches 12 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 42.7°

b. 22.6°

c. 72.4°

d. 67.4°

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Kinematics

A kite string 50 m long is pulled taut, and the kite is 32 m above the hand holding it. Find the angle of elevation of the string, to the nearest tenth of a degree.

a. 50.2°

b. 39.8°

c. 32.6°

d. 44.8°

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 12.01 Forces & Equilibrium

The function $f(x) = (x + 6)^2 - 3$ has domain $x \geq -6$. State the range of f.

a. $f(x) \leq -3$

b. $f(x) \geq -3$

c. $f(x) \geq -6$

d. $x \geq -6$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 13.01 Newton's Laws

A 15 m rope is tied to the top of a pole and staked to level ground 9 m from the base of the pole. Find the angle the rope makes with the ground, to the nearest tenth of a degree.

a. 36.9°

b. 31.0°

c. 58.1°

d. 53.1°

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 14.01 Energy, Work & Power

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5°

b. 81.5°

c. 13.5°

d. 17.1°

© ClipSAT — clipsat.org | All Rights Reserved